DPPC: Discriminative Pairwise Predictive Correction with Comparative Prompts for Training-Free Few-Shot Class-Incremental Learning

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Abstract

Few-shot class-incremental learning (FSCIL) must recognize all seen classes while estimating each arriving class from only a few labeled examples. Fixed-parameter vision–language scorers avoid repeated optimization, but their classwise scores may omit attributes that distinguish two plausible labels. We introduce Discriminative Pairwise Predictive Correction (DPPC), a no-gradient auxiliary reranker that leaves the scorer rule and representation unchanged. Within the scorer's top-K candidates, DPPC measures each novel-touching pair along comparative-prompt and class-name directions. Legal class supports define one-dimensional posterior-predictive contrasts; a weak base-derived variance prior keeps them defined at one shot, and empirical separation weights attenuate poorly supported directions. A bounded, antisymmetric correction preserves the scorer's native scale and every shortlisted base–base score gap. Only K and correction strength are selected, using labeled-base pseudo-sessions with disjoint fit/query rows before incremental evaluation. Across six directly implemented scorers on CIFAR-100, CUB-200, and miniImageNet, DPPC improves mean five-shot session-average and final accuracy in all 18 matched cells. Session-average gains range from $+0.13$ to $+3.81$ points over 20 paired support manifests; every paired interval is positive and every exact sign test survives Holm correction. Support–class resampling retains positive intervals in 16/18 cells, and crossed order–support resampling does so in 18/18. Benefits contract or reverse in several high-shot CUB-200 settings, and base accuracy can fall while novel accuracy rises. DPPC is therefore a bounded correction for residual low-shot ambiguity, not a universal accuracy or calibration guarantee.

1. Introduction

Few-shot class-incremental learning (FSCIL) begins with a many-shot base session and then introduces disjoint classes from only a few labeled examples [21, 22]. After each arrival, one classifier must recognize every class seen so far. New-class estimates supported by only a few images must therefore compete with better-estimated base classes without destroying prior recognition [33].

Frozen vision–language features make this setting attractive because class names and generated descriptions supply semantic priors while labeled supports provide task-specific visual evidence [14, 17, 18, 31]. Here a *fixed-parameter scorer* has a fixed representation and scoring rule but may legally update support-derived prototypes or sufficient statistics as classes arrive. Bi-Level Modality Calibration (BiMC), for example, combines text, prototypes, covariance, and description retrieval without gradient updates [1]. Such a scorer produces a margin for every class pair, yet the evidence assigned to each endpoint is normally constructed classwise. The margin may consequently omit an attribute that specifically distinguishes its two endpoints.

Rival-conditioned language can resolve ambiguity in static recognition, as shown by FuDD and Comparative Descriptors [2, 10]. Comparative prompting itself is therefore prior art. The unresolved FSCIL question is whether a textual direction separates the labeled examples currently available for the two classes. Text-space contrast or query–text affinity alone does not supply that class-conditional support test.

We introduce *Discriminative Pairwise Predictive Correction* (DPPC), illustrated in Fig. 1. Within the unchanged scorer's top-K candidates, DPPC forms an antisymmetric direction from separately generated prompt orders and a second direction from the class-name template ensemble. Session-available supports yield one-dimensional posterior-predictive evidence along each direction. A weak base-derived variance prior keeps this evidence defined at one shot, while empirical separation weights attenuate axes that do not separate their endpoints. DPPC fuses the two branches,

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retains only pairs touching an incremental class, and projects the resulting zero-sum graph correction so every shortlisted base class receives the same shift. The rule therefore preserves all shortlisted base–base score gaps, and its bounded evidence limits any corrected pairwise-gap change to 2η.

No encoder, scorer parameter, prompt embedding, or session statistic is updated by gradient descent. DPPC nevertheless uses supervision: it consumes the frozen query feature, legal labeled class features or moments, text axes, and a base/incremental mask. Six labeled-base pseudo-streams, each with disjoint fit/query rows, select only shortlist width K and strength η before incremental evaluation; they see neither target-session supports nor evaluation-query labels or scores. The reported benchmarks expose the complete class-name ontology when the prompt bank is constructed, and all comparative prompts come from one provider.

The evaluation pairs DPPC with six directly implemented scorers on three benchmarks and 20 matched support manifests. It also rebuilds representations, varies support size and class order, compares matched scalar-bias and ULA controls, and audits decisions at query level. The resulting evidence distinguishes aggregate low-shot gains from universal retention: high-shot, base-class, and individual-query harms are retained.

Our contributions are:

- a support-validated semantic measurement that turns comparative and class-name directions into bounded finite-support predictive evidence;
- a native-scale, novel-touching correction with exact shortlisted base–base-gap invariance and base-only selection of its two operating quantities; and
- a matched evaluation across score families, representations, support sizes, class orders, and negative controls that reports both gains and failure boundaries.

2. Related work

FSCIL representations and prompts. Early FSCIL methods learn the classifier, representation, or their geometry: CEC evolves classifiers, FACT prepares forward-compatible features, ALICE uses an open-set objective, SAVC imposes semantic virtual constraints, and NC-FSCIL aligns features with the classifier [16, 20, 29, 30, 32]. More recent methods adapt pretrained vision–language models. PriViLege tunes pretrained transformers; SEC-Prompt, QR-Prompt, and SGLC learn prompt modules; and DSC optimizes a semantic-differential visual representation before prototype calibration [7, 12, 15, 19, 28]. DPPC instead retains the chosen representation and scorer rule and modifies only the final decision among a short list.

Fixed-parameter scoring and post-hoc correction. TEEN transfers base-class relations to novel prototypes [25]; FeCAM and RanPAC show the strength of covariance and random-projection classifiers on frozen features [4, 13]. BiMC combines class text, prototypes, covariance, and retrieved descriptions, FDR decomposes frozen vision–language features, and HyCal mixes cosine and regularized Mahalanobis geometry [1, 9, 27]. ADBS learns class-specific boundaries, whereas ULA post-hoc adjusts FSCIL logits using groupwise margins and base-entropy uncertainty [11, 26]. Classical temperature scaling calibrates confidence without changing the predicted class [5]; DPPC instead makes a bounded local ranking change and does not claim globally calibrated probabilities.

Pairwise coupling combines one-vs-one probability estimates into a coherent multiclass distribution [6]. DPPC has a different interface: it neither trains binary classifiers nor couples normalized pair posteriors. It uses standard one-dimensional posterior-predictive densities to measure support compatibility, antisymmetrizes the resulting contrasts, and applies their constrained graph divergence only to novel-touching shortlist edges. The contribution lies in this FSCIL interface, support validation, and base-gap-preserving correction rather than in the conjugate Student-t model alone.

Language-derived comparative evidence. CLIP enables recognition from class-name prompts, while generated descriptions enrich classwise visual evidence [14, 17, 18]. BPC-CLIP combines fine-grained descriptions with gated visual–textual prompt fusion [3]. FuDD generates query-conditioned differential descriptions, and Comparative Descriptors form pair-relative descriptions and filter them in image space for static recognition [2, 10]. DPPC does not claim comparative language itself. Its distinction is to use a frozen bidirectional prompt bank, test both comparative and class-name directions against legal class supports, model their finite-support uncertainty, and re-rank an unchanged FSCIL scorer with K and η selected only from labeled base data.

3. Method

DPPC is a no-gradient, base-calibrated auxiliary re-ranker. For a query it takes the fixed-parameter scorer row $s^0(x)$ *and* a frozen visual feature z_x; its session state contains legal labeled class features (or their sufficient moments), two frozen text-axis banks, and the base/incremental class mask. It does not modify or backpropagate through the input scorer. The scorer may update its prescribed support-derived state as classes arrive; DPPC changes neither that rule nor its state. Thus the intervention preserves the scorer but is neither score-only nor label-free.

The fixed rule comprises dual semantic measurements, a finite-support predictive model, empirical separation weighting, and a constrained local graph correction. Only shortlist width K and nonnegative strength η are fit, separately for

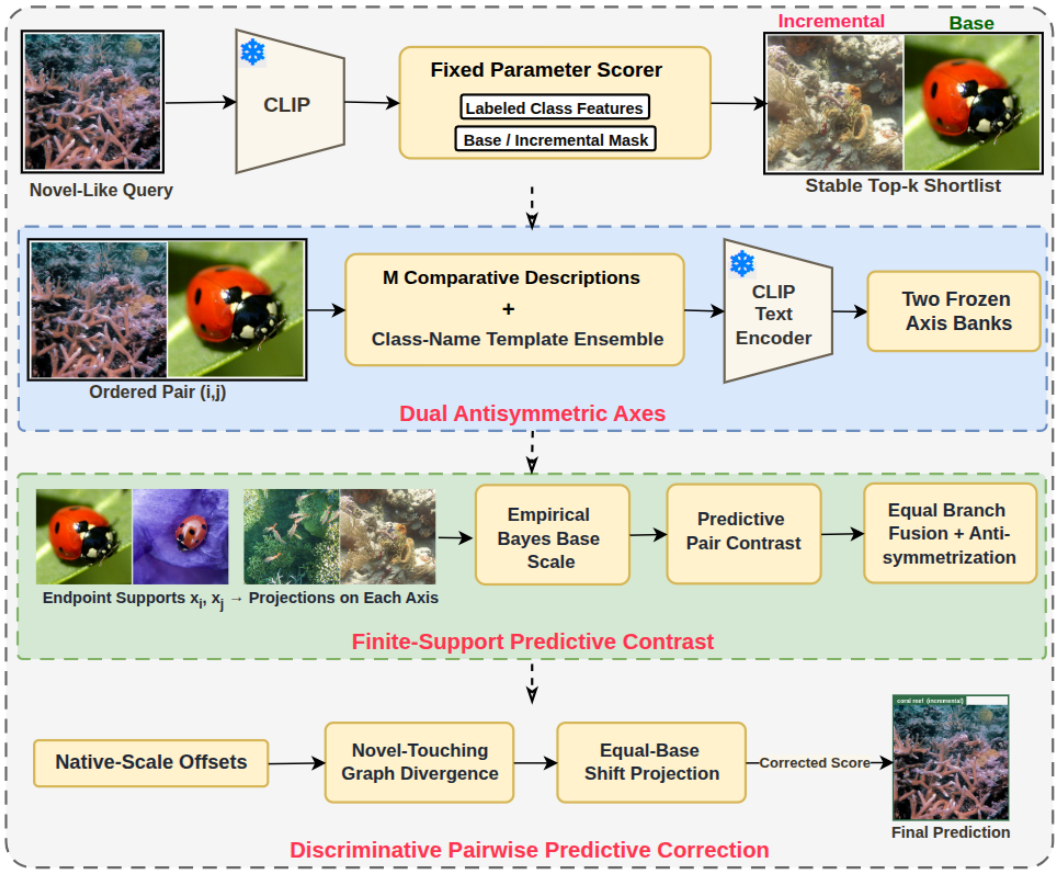


Figure 1. DPPC leaves the scorer call unchanged and separately consumes the frozen query feature. Legal class features determine pairwise predictive contrasts and separation weights. Labeled-base pseudo-sessions with disjoint fit/query rows select only (K, η). The selector uses neither actual incremental-session supports nor evaluation-query labels or scores.

each scorer–dataset setting, from labeled base data. The prompt contract and member count, prior degree, equal branch fusion, evidence bound, and novel-touching graph are disclosed fixed design choices rather than learned quantities. The canonical five-shot study derives them separately for each scorer–dataset pair; shot-transfer controls either retain that pair or repeat the same base-only derivation at the new shot count.

3.1. FSCIL state and interface

An FSCIL stream contains disjoint class sets $\mathcal{C}_0, \ldots, \mathcal{C}_T$. Session 0 provides many labeled examples for base classes; session $t > 0$ provides a few labeled supports for new classes in $\mathcal{C}_t$. Evaluation covers $\mathcal{C}_{\le t} = \cup_{r=0}^{t} \mathcal{C}_r$, and every class in $\mathcal{C}_{\le t} \setminus \mathcal{C}_0$ is called incremental below, including classes introduced before session t. Let frozen image and text encoders be E_I, E_T, define $\mathrm{nrm}(0) = 0$ and $\mathrm{nrm}(v) = v/\|v\|_2$ otherwise, and set $z_x = \mathrm{nrm}(E_I(x)) \in \mathbb{R}^d$. The set $\mathcal{X}_i^t$ contains only the labeled features legally available for class i at session t; its moments may be cached after arrival. Write $\mathcal{S}_{\le t} = \{\mathcal{X}_i^t : i \in \mathcal{C}_{\le t}\}$ for this legal support state and θ_A for fixed scorer parameters and constants. A higher-is-better scorer returns

$$s^0(x) = A(z_x, \mathcal{S}_{\le t}; \theta_A). \tag{1}$$

DPPC changes neither A nor the prototypes, temperatures, covariance state, or fusion rules that A constructs from $\mathcal{S}_{\le t}$. The reported scorers and DPPC share a frozen CLIP feature space; applying the interface with a different representation requires rebuilding every representation-dependent DPPC asset.

3.2. Dual antisymmetric semantic axes

For every ordered pair (i, j), the fixed bank stores M descriptions $p_{ij}^{(m)}$ generated for i relative to j separately from the reverse order. We encode both directed sets:

$$h_{ij} = \mathrm{nrm}\left(\frac{1}{M}\sum_m \mathrm{nrm}(E_T(p_{ij}^{(m)}))\right),$$
$$a_{ij} = h_{ij} - h_{ji} = -a_{ji}. \tag{2}$$

We do not renormalize a_{ij}: doing so would manufacture a unit direction when the two orders nearly coincide. Our definition of nrm also makes the zero-mean prompt case explicit. A second axis $a'_{ij} = t_i - t_j$ uses $t_i = \mathrm{nrm}(|\mathcal{T}|^{-1} \sum_{\tau \in \mathcal{T}} \mathrm{nrm}(E_T(\tau(\mathrm{name}_i))))$, where $\mathcal{T}$ is the dataset manager's fixed CLIP template ensemble. The generator receives class names and the fixed ontology, but no image, support set, prediction, or performance value. The reported bank uses one provider and one disclosed contract, so provider transfer is not assumed.

3.3. Finite-support predictive contrast

We suppress the session superscript below and apply the following model independently to a_{ij} and a'_{ij}. Write $y_{ij}(z) = z^\top a_{ij}$ and $n_c = |\mathcal{X}_c^t|$. Conditional on one shared pair variance, endpoint projections are modeled as $y_{ij}(z) \mid \mu_{c|ij}, \sigma_{ij}^2 \sim \mathcal{N}(\mu_{c|ij}, \sigma_{ij}^2)$, with a flat prior on each endpoint mean. Under that flat prior, the posterior location $\mu_{c|ij}$ is the legal-support sample mean; let $\widehat{v}_{c|ij}$ denote its population-form variance. To avoid a zero residual degree at one shot, we estimate an isotropic empirical-Bayes scale from labeled base data,

$$\begin{aligned} \widehat{\sigma}_B^2 &= \frac{\sum_{b \in \mathcal{C}_0} \sum_{z \in \mathcal{X}_b^0} \|z - \bar{z}_b\|_2^2}{d \sum_{b \in \mathcal{C}_0} (n_b - 1)}, \\ v_{ij}^0 &= \max(\widehat{\sigma}_B^2 \|a_{ij}\|_2^2, \epsilon), \qquad \epsilon = 10^{-12}. \end{aligned} \tag{3}$$

The denominator is positive in the stated protocol because every base and pseudo-base class has multiple rows. We place the scaled inverse-χ^2 prior $\sigma_{ij}^2 \sim \mathrm{Inv}\text{-}\chi^2(\lambda, v_{ij}^0)$ with the fixed weak degree $\lambda = 1$. Above the declared numerical floor, the axis factor makes the model invariant to a common nonzero rescaling of the axis. The isotropic base scale is a shrinkage device, not a claim that CLIP residuals are isotropic; endpoint data retain their observed axis-specific variances. Conjugate integration gives

$$\begin{aligned} \nu_{ij} &= n_i + n_j - 2 + \lambda, \\ s_{ij}^2 &= \frac{n_i \widehat{v}_{i|ij} + n_j \widehat{v}_{j|ij} + \lambda v_{ij}^0}{\nu_{ij}}, \\ q_{c|ij}(x) &= t_{\nu_{ij}}\left(y_{ij}(z_x); \mu_{c|ij}, s_{ij}^2(1 + n_c^{-1})\right), \\ e_{ij}(x) &= \tanh\left[\frac{1}{2}\{\log q_{i|ij}(x) - \log q_{j|ij}(x)\}\right]. \end{aligned} \tag{4}$$

Here $t_\nu(y; \mu, \tau^2)$ denotes the Student-t density with squared scale τ^2. With equal priors on the two endpoints, $e_{ij} = (q_{i|ij} - q_{j|ij})/(q_{i|ij} + q_{j|ij}) \in [-1, 1]$; it is a bounded pairwise contrast, not a full-class posterior. In particular, a one-shot pair uses a defined t_1 predictive rather than a zero-variance plug-in density.

We shrink each branch by the observed separation relative to estimated mean uncertainty,

$$\rho_{ij} = \frac{(\mu_{i|ij} - \mu_{j|ij})^2}{(\mu_{i|ij} - \mu_{j|ij})^2 + s_{ij}^2(n_i^{-1} + n_j^{-1})} \in [0, 1]. \tag{5}$$

This is an empirical signal-to-noise weight, not a calibrated probability that the axis is correct; noisy supports can still assign a large weight to a wrong direction. Primed quantities denote the class-name branch. The fixed equal-weight fusion and explicit antisymmetrization are

$$\begin{aligned} f_{ij}(x) &= \tfrac{1}{2}(\rho_{ij} e_{ij}(x) + \rho'_{ij} e'_{ij}(x)), \\ \widetilde{e}_{ij}(x) &= \tfrac{1}{2}(f_{ij}(x) - f_{ji}(x)). \end{aligned} \tag{6}$$

Hence $\widetilde{e}_{ji}(x) = -\widetilde{e}_{ij}(x)$ and $|\widetilde{e}_{ij}(x)| \le 1$.

3.4. Native-scale correction and base–base invariance

The scorer's stable top $\bar{K}_t = \min(K, |\mathcal{C}_{\le t}|)$ labels $c_1, \ldots, c_{\bar{K}_t}$ define the ambiguity set. We retain the scorer's native scale, without query-wise standardization, by subtracting only a common constant:

$$u_k(x) = s_{c_k}^0(x) - \max_\ell s_{c_\ell}^0(x). \tag{7}$$

This operation preserves all pre-correction gaps; DPPC may subsequently change novel-touching gaps. A stable descending sort retains class-column order among equal raw scores, and a corrected-score tie prefers that anchor order. Consequently η is scorer-specific and expressed in the scorer's units. We restrict $\eta \ge 0$ because the likelihood ratio fixes the semantic orientation; a negative value would reverse every pair measurement rather than attenuate it, for which the admitted fallback is $\eta = 0$.

Let $r_{k\ell} = 1$ when at least one endpoint is incremental and zero otherwise. The scaled divergence of the antisymmetric shortlist graph is defined piecewise so a degenerate one-class interface is also total:

$$g_k(x) = \begin{cases} 0, & \bar{K}_t = 1, \\ \dfrac{1}{\bar{K}_t - 1} \displaystyle\sum_{\ell \ne k} r_{k\ell} \widetilde{e}_{c_k c_\ell}(x), & \bar{K}_t > 1. \end{cases} \tag{8}$$

The denominator is deliberately the complete-shortlist degree rather than the number of active edges: evidence shrinks when a shortlist contains few incremental candidates. Every active edge contributes equal and opposite mass, so $\sum_k g_k(x) = 0$. Let $\mathcal{B}_x = \{k : c_k \in \mathcal{C}_0\}$. The Euclidean projection that leaves incremental coordinates unchanged and constrains all shortlisted base shifts to be equal is

$$\begin{aligned} D_k(x) &= \begin{cases} |\mathcal{B}_x|^{-1} \sum_{r \in \mathcal{B}_x} g_r(x), & k \in \mathcal{B}_x, \\ g_k(x), & k \notin \mathcal{B}_x, \end{cases} \\ s_k^D(x) &= u_k(x) + \eta D_k(x), \qquad \widehat{y} = c_{\arg\max_k s_k^D(x)}. \end{aligned} \tag{9}$$

If $\mathcal{B}_x$ is empty, $D = g$; when $\bar{K}_t = 1$, this gives $D = 0$. Projection preserves the graph sum and keeps $D_k \in [-1, 1]$. All shortlisted base candidates receive the same shift, so base–base gaps are exactly preserved, although base–incremental

gaps and base accuracy need not be. In session 0 no edge is active and the complete prediction is unchanged. A truth outside the shortlist remains unrecoverable. Any corrected gap changes by at most 2η; this is a reachability bound, not a guarantee that the change has the correct sign.

3.5. Base-supervised, evaluation-isolated selection

For each canonical scorer–dataset pair, six deterministic pseudo-FSCIL episodes split the dataset's labeled base classes into pseudo-base and pseudo-incremental groups, with disjoint statistic/support and pseudo-query rows. Thus the selector intentionally uses base labels and base scorer outputs. It receives no actual incremental-session support, evaluation-query label, or evaluation-session score row. Six episodes are a fixed design choice rather than an inferred sample size; the supplement audits 13 deterministic selector constructions without substituting an alternative chosen from target results.

We evaluate the fixed candidate set $K \in \{2, 5, 10, 15, 20, 30, 40\}$ using session- and class-balanced accuracy in $[0, 1]$. For fixed K, every corrected candidate score is affine in $\eta \geq 0$. The strengths for which a query's true shortlisted candidate wins form an interval, so we intersect the pairwise inequalities and sweep all open regions. We retain $\eta = 0$ and only open regions containing a value representable in the deployed score dtype; positive endpoint-only ties and sub-lattice regions are excluded. The deployed arithmetic must reproduce the complete predicted-correctness vector of the selected region. Within 10^{-6} accuracy of the optimum, ties favor the smaller K and then the smallest deployment-representable η. This is an exact one-dimensional numerical maximization for the fixed pseudo-episodes, not a statistical generalization guarantee.

The selected pair is frozen before all incremental-evaluation support resamplings. The reported benchmarks expose the full class ontology when the prompt bank is built. If future names are hidden, axes must instead be built at arrival and base features or covariance sufficient statistics retained. A sparse implementation can project only the $K(K-1)$ gathered axes in $O(K^2 d)$ per query, whereas the timed reference kernel projects the dense seen-class bank in $O(|\mathcal{C}_{\leq t}|^2 d)$ before gathering. Prompt generation has $O(|\mathcal{C}|^2)$ directed outputs, and their encoding/storage costs $O(|\mathcal{C}|^2 d)$; reported timing is correction-only.

Algorithm 1 DPPC session setup and query-time correction

Require: $s^0(x), z_x$; frozen axes; legal class moments; base mask; fixed (K, η)

1: Form the stable top-K shortlist and native-scale offsets $u(x)$
2: Evaluate both posterior-predictive pair contrasts from z_x
3: Apply empirical separation weights; fuse and antisymmetrize
4: Take fixed-denominator graph divergence; project base shifts to equality
5: Return $\arg\max_k [u_k(x) + \eta D_k(x)]$

4. Experimental protocol

Benchmarks and evaluation rows. CIFAR-100 and miniImageNet use 60 base classes followed by eight five-class sessions; CUB-200 uses 100 base classes followed by ten ten-class sessions [8, 23, 24]. Base classes use all legally available training rows. For each of 20 deterministic manifests, five training-pool supports are sampled independently for every arriving class; both arms receive the identical manifest. Evaluation uses all fixed test rows of the classes visible after each session, totaling 72,000, 47,538, and 72,000 session-level query predictions per manifest on CIFAR-100, CUB-200, and miniImageNet. The canonical class order is fixed; a separate audit completely rebuilds all class-indexed assets for ten hash-defined alternative orders.

Representation, prompts, and ontology. The primary study uses OpenAI CLIP ViT-B/16 with its released image transform and the dataset manager's class-name template ensemble [18]. Each ordered class pair has three distinct comparative descriptions, all encoded by the same frozen CLIP text encoder. One disclosed generator and contract supply the entire bank. Prompt generation receives the class names and complete benchmark ontology but no image, support, score, sample–label assignment, or performance value. Thus the canonical deployment assumes future class names are known when the bank is built; arrival-time generation is required otherwise.

Direct scorer family. We pair DPPC with six higher-is-better scorer definitions: CLIP zero-shot, cosine nearest-class-mean (NCM), TEEN [25], shared-covariance linear discriminant analysis (LDA), and the official BiMC-Sgl and BiMC-Ens inference paths [1]. These $6 \times 3 = 18$ released or analytic cells form the primary family. FDR and HyCal could not be evaluated with official source parity; clearly marked reconstructions and their conventions are retained only as supplementary transfer evidence [9, 27]. This paired design tests the incremental value of DPPC for a fixed score

definition, not a ranking of differently trained full systems.

Selection and pairing. For each scorer–dataset pair, six deterministic labeled-base pseudo-streams use disjoint statistic/support and query rows to select (K, η). They receive no actual incremental support, evaluation-query label, or target-session score row. The choice is then fixed across the 20 target support manifests. Every changed representation, shot-matched protocol, prompt subset, or semantic control receives its own base-only derivation; target performance never chooses between operating points.

Metrics and inference. The primary metric is $\mathrm{Avg} = (T+1)^{-1} \sum_{t=0}^{T} \mathrm{Acc}_t$, including session 0. Final overall, final base/novel, and the mean incremental-session base–novel harmonic accuracy H_{inc} are secondary. Paired Student-t 95% intervals vary the support manifest on fixed queries and order. For five-shot Avg, exact paired sign tests receive Holm correction across the 18 direct cells; the six reconstruction cells form a separate family. Fixed 50,000-replicate bootstraps additionally resample support manifests with base/arrival-stratified query-class clusters and the complete 11×20 order–support matrix. These are secondary robustness analyses: they do not alter DPPC, (K, η), or any target prediction. Only Avg receives family-wise efficacy testing, and unfavorable secondary results are retained. The three semantic controls use one joint Holm family over their 54 full–control Avg contrasts.

Matched controls. An exact base-only breakpoint sweep selects one scorer-specific scalar shift for all incremental-class scores; its target comparison uses the same 20 manifests. A released ULA inference rule is also evaluated on matched cosine-NCM logits [26]. The supplement further reports semantic-branch, prompt assignment, component, corruption, representation, shot, and order controls.

5. Results

5.1. Fixed-parameter scorer comparison

Table 1 pairs every direct primary scorer with the identical scorer plus DPPC at 1, 5, 10, and 20 shots. At five shots, all 18 direct scorer–dataset pairs have higher mean Avg and Final accuracy. Avg gains range from $+0.125$ to $+3.807$ points, and every support-seed paired 95% interval excludes zero. Sixteen pairs improve on all 20 manifests. The exceptions are CIFAR-100 BiMC-Sgl, with $16/0/4$ wins/ties/losses and $+0.125$ $[+0.067, +0.184]$, and CUB-200 BiMC-Ens, with $17/0/3$ and $+0.175$ $[+0.079, +0.272]$. All exact sign tests survive Holm correction (largest adjusted $p = 1.18 \times 10^{-2}$). Support–class intervals remain positive in 16/18 cells, with the same two small-effect cells crossing zero; crossed order–support intervals are positive in 18/18. These analyses vary observed benchmark units rather than supplying external datasets.

The gain is not uniform retention. Final-base accuracy is positive in only 6/18 cells, whereas H_{inc} has a positive interval in 17/18. Across shot-matched 1, 5, 10, and 20-shot settings, 68/72 Avg intervals are positive; the four reversals are high-shot CUB-200 cells. As support-based scorers improve, DPPC's residual gain generally contracts. Full trajectories, operating points, marked FDR/HyCal reconstructions, and all secondary intervals are reported in the supplement.

6. Ablation studies

Panels (a–h) of Fig. 2 show CIFAR-100 diagnostics; panel (i) compares all 18 direct cells. Full CUB-200 and mini-ImageNet results, complete intervals, and unfavorable cells appear in the supplement.

Panel-level interpretation. **(a)** Gain is zero in the all-base session by construction and generally rises as arrivals create novel-touching edges. Different slopes indicate scorer-specific residual confusion, not a common score shift. **(b)** BiMC-Ens gain contracts as additional support stabilizes its prototypes and covariance estimates, leaving less error for semantic re-ranking. **(c)** K trades omission against noise: small lists miss plausible competitors; large ones add $O(K^2)$ unrelated uncertain edges that can override native margins.

(d) Zero strength exactly recovers the scorer; the rise and subsequent fall show a useful correction plateau followed by over-correction. The target peak near 0.75η is diagnostic and does not replace the base-derived η. **(e)** Arrival order changes which confusable pairs coexist and how long they contribute to Avg, explaining the spread, while every CIFAR-100 interval remains positive; the benchmark order is therefore not the sole source of gain. **(f)** A positive corrupted-support contrast is relative to the same corrupted scorer, not evidence that corruption helps absolutely. Drops and duplicates can create more correctable scorer errors, whereas cross-class and base-class mismatches contaminate class moments and shrink the gain; bounded antipodal evidence stays near the clean contrast.

(g) Every prompt subset helps, but the absence of a monotone member-count trend shows that descriptions are partly redundant rather than interchangeable; averaging can either reinforce or dilute pair-specific signal. **(h)** A single base-selected bias captures some global base–novel skew, especially for LDA and BiMC-Ens, but the remaining DPPC–bias contrasts require query- and pair-specific evidence. **(i)** Most cells exchange some final-base accuracy for a larger harmonic gain because DPPC modifies novel-touching decisions while guaranteeing only base–base gap preservation; the method improves base–novel balance, not uniform retention.

Structural controls. Each arm in Tab. 2 replaces one design

Table 1. Direct-scorer session-average accuracy on matched CLIP ViT-B/16 features (%, 20 paired support manifests). DPPC additionally consumes the frozen query/support features and text axes. Each block shows the scorer, its +DPPC result, and a shaded bold Δ row; bold absolute accuracy marks the highest observed value across unchanged and corrected rows for that dataset and shot. Shot-matched labeled-base pseudo-sessions select (K, η) before incremental evaluation; five shots is canonical. Paired intervals, complete metrics, and clearly marked reconstruction/control results are supplementary.

	CIFAR-100				CUB-200				miniImageNet			
Method / support shots	1	5	10	20	1	5	10	20	1	5	10	20
CLIP zero-shot, *ICML'21* [18]	70.86	70.86	70.86	70.86	58.04	58.04	58.04	58.04	88.69	88.69	88.69	88.69
+ DPPC, *ours*	71.33	71.58	71.66	71.72	59.05	59.77	59.86	60.02	89.49	89.85	89.89	89.92
Δ	**+0.47**	**+0.72**	**+0.80**	**+0.86**	**+1.01**	**+1.73**	**+1.82**	**+1.98**	**+0.80**	**+1.16**	**+1.20**	**+1.23**
Cosine NCM, *analytic*	60.81	67.87	70.00	71.14	61.66	73.09	75.93	77.40	79.71	88.94	89.94	90.31
+ DPPC, *ours*	63.87	70.79	71.78	72.60	66.75	75.15	76.85	78.07	85.72	90.46	90.87	91.12
Δ	**+3.05**	**+2.93**	**+1.78**	**+1.46**	**+5.09**	**+2.06**	**+0.93**	**+0.67**	**+6.01**	**+1.53**	**+0.93**	**+0.80**
TEEN, *NeurIPS'23* [25]	62.99	67.73	68.88	69.52	65.06	71.95	74.03	75.29	84.93	89.12	89.56	89.71
+ DPPC, *ours*	66.87	71.54	72.32	72.72	69.11	74.06	76.05	76.56	87.90	90.73	91.01	91.13
Δ	**+3.88**	**+3.81**	**+3.43**	**+3.20**	**+4.06**	**+2.11**	**+2.02**	**+1.27**	**+2.97**	**+1.62**	**+1.45**	**+1.42**
Shared LDA, *analytic*	65.49	73.79	76.24	77.43	62.28	75.99	79.63	**81.34**	80.98	90.84	92.14	92.70
+ DPPC, *ours*	68.87	76.51	**77.59**	**78.06**	68.45	**78.22**	**80.37**	81.23	86.57	92.46	93.03	93.20
Δ	**+3.38**	**+2.72**	**+1.35**	**+0.63**	**+6.18**	**+2.24**	**+0.73**	**−0.11**	**+5.59**	**+1.63**	**+0.88**	**+0.50**
BiMC-Sgl, *CVPR'25* [1]	72.51	75.58	75.90	76.08	67.54	75.69	77.32	78.17	90.97	93.17	93.41	93.54
+ DPPC, *ours*	73.93	75.71	76.05	76.18	69.80	76.22	77.39	78.14	92.24	93.38	93.54	93.65
Δ	**+1.43**	**+0.13**	**+0.15**	**+0.09**	**+2.27**	**+0.54**	**+0.07**	**−0.03**	**+1.27**	**+0.20**	**+0.13**	**+0.11**
BiMC-Ens, *CVPR'25* [1]	69.13	75.43	76.26	76.68	69.15	76.31	77.14	77.33	90.10	93.22	93.54	93.69
+ DPPC, *ours*	**74.70**	**77.29**	77.57	77.67	**71.36**	76.48	76.83	77.11	**93.24**	**93.66**	**93.79**	**93.86**
Δ	**+5.56**	**+1.86**	**+1.31**	**+0.99**	**+2.20**	**+0.18**	**−0.31**	**−0.22**	**+3.14**	**+0.43**	**+0.25**	**+0.17**

bundle and independently re-derives (K, η). Query standardization erases the native margin scale on which η operates; removing the variance prior makes sparse class moments easier to over-trust; relative-only fusion replaces absolute branch attenuation with weighting by relative separation; and row means remove the zero-sum/base-gap constraints. Full DPPC beats these controls in 17, 13, 7, and 4 of 18 mean-Avg cells, respectively. The last two counts also show that the constraints encode stable behavior rather than target-wise optimality.

Semantic specificity. With every arm independently base-derived in same-cache matched runs, comparative-only correction improves 18/18 mean Avg cells (17 positive intervals), while class-name-only improves 16/18 (16 positive intervals). Full DPPC exceeds a cyclically mismatched comparative bank in all 18 intervals; 16 exact sign tests survive one Holm family over all 54 semantic-control contrasts. Fusion is not uniformly best: full exceeds comparative-only in 14/18 intervals and class-name-only in 13/18, while class-name-only beats full for CUB-200 TEEN by 0.341 $[0.219, 0.463]$ point. Thus correct pair assignment matters, but neither branch is universally necessary.

Table 2. Structural controls with BiMC-Ens on CIFAR-100 at five shots (%, 20 support manifests). R: native scorer scale; V: base-derived variance-prior bundle; A: empirical separation weighting; G: constrained zero-sum/base-gap aggregation bundle. Every control independently re-derives (K, η). Bold marks the best observed value; shading identifies full DPPC.

Configuration	R	V	A	G	Avg	Novel	H_{inc}
Unchanged					75.43	52.34	65.73
Query z-score		✓	✓	✓	77.14	64.26	72.63
No variance prior	✓		✓	✓	77.30	71.84	75.07
Relative fusion	✓	✓		✓	**77.39**	71.48	74.97
Row mean	✓	✓	✓		77.17	68.76	73.82
Full DPPC	✓	✓	✓	✓	77.29	**71.89**	**75.09**

Scope beyond the panels. Across all datasets, the base-selected scalar bias helps 10/18 cells, harms two, and is inconclusive in six; DPPC exceeds it in 18/18 after Holm correction. DPPC changes 1.41M of 22.98M accumulated

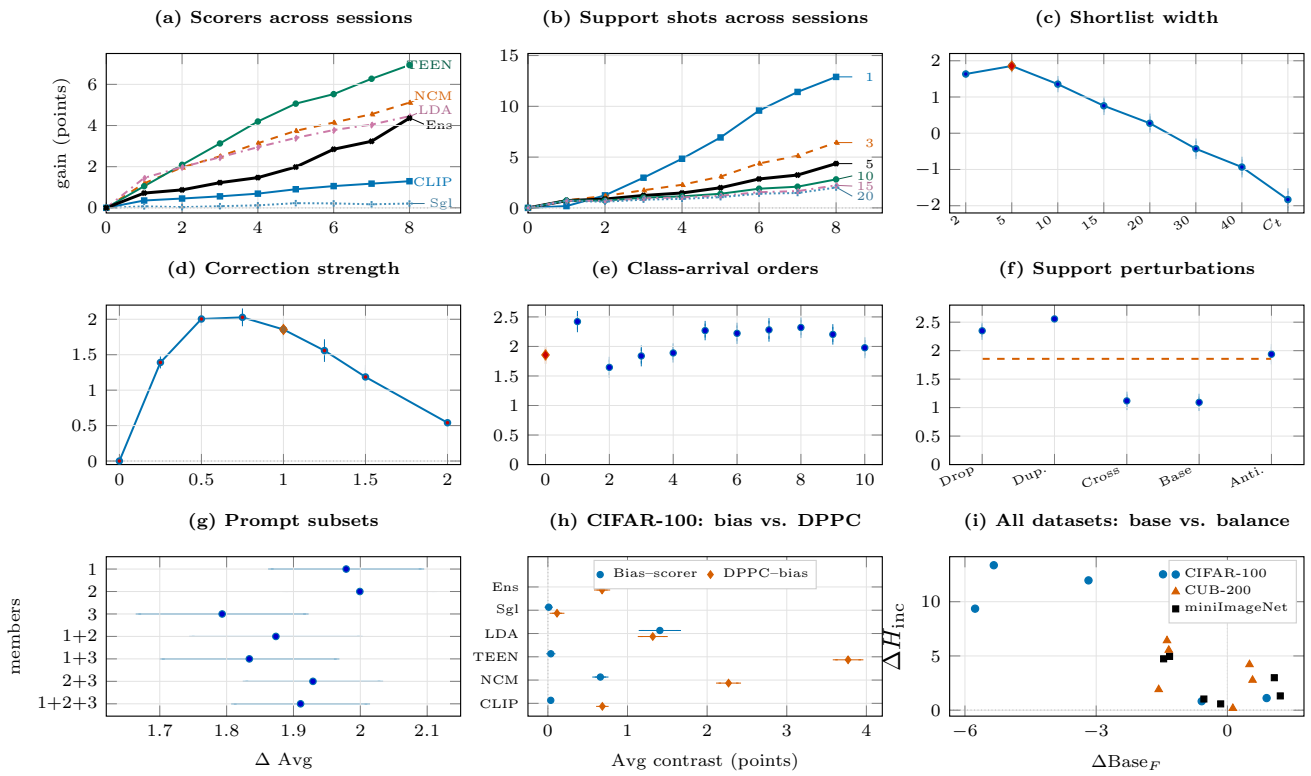


Figure 2. Matched diagnostics. (a) Scorer-wise session gains; (b) BiMC-Ens across support sizes; target-only (c) K and (d) strength sweeps; (e) class orders; (f) support perturbations; (g) prompt subsets; (h) scalar-bias contrasts; and (i) final-base versus H_{inc} changes in all 18 cells. Bars in (c–h) are paired 95% intervals. In (h), blue circles are bias–scorer and orange diamonds DPPC–bias. In (i), color/shape identifies CIFAR-100 (blue circle), CUB-200 (orange triangle), and miniImageNet (black square). Orange diamonds in (c–e) mark the base-only choice or benchmark order; the dashed line in (f) is clean support.

predictions (0.66M repairs, 0.27M harms, and 0.47M wrong-to-wrong changes), so it is not an error detector. Complete RN101, ViT-B/32, ViT-B/16, and ViT-L/14 rebuilds yield 24/24 positive miniImageNet scorer–encoder intervals, establishing representation transfer within this tested family rather than universality.

7. Discussion and limitations

DPPC is a local decision correction, not a probability-calibration method or a replacement scorer. It preserves scorer parameters and rules, not session state: NCM, LDA, TEEN, and BiMC still form their legal support statistics, while DPPC additionally needs compatible frozen features, class moments, and a base/incremental mask. Its invariants do not guarantee accuracy: aggregate gains coexist with base-class, individual-query, and calibration harms; zero-support DPPC is undefined; and several high-shot CUB-200 effects reverse.

The evidence covers six direct score definitions, three benchmarks, and four miniImageNet encoders, but is limited to one prompt provider with imperfect contract compliance and a known benchmark ontology; FDR and HyCal are reconstructions rather than official-parity implementations. Hidden future names require arrival-time generation and retained base statistics. Reported timing is correction-only and excludes prompt generation, encoding, scorer latency, and the offline $O(|\mathcal{C}|^2 d)$ bank. Provider transfer, an external method-matched benchmark, and a sparse end-to-end implementation remain open tests.

8. Conclusion

DPPC converts support-validated comparative evidence into a bounded correction of an unchanged fixed-parameter FSCIL scorer. Matched gains across score families show useful residual low-shot correction, while base, query, prompt, and high-shot failures delimit the claim. The method is best viewed as a local decision repair when legal supports validate a semantic distinction, not as a general guarantee of retention or calibration.

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